

Accelerating kinetic plasma simulations with machine-learning-generated initial conditions

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ABSTRACT

Computational models of plasma technologies often solve for the system operating conditions by time-stepping an initial value problem to a quasi-steady solution. However, the strongly nonlinear and multi-timescale nature of plasma dynamics often necessitate millions, or even hundreds of millions, of steps to reach convergence, reducing the effectiveness of these simulations for computer-aided engineering. We consider acceleration of kinetic plasma simulations via data-driven machine-learning-generated initial conditions, which initialize the simulations close to their final quasi-steady-state, thereby reducing the number of steps to reach convergence. Three machine-learning models are developed to predict the density and ion kinetic profiles of capacitively coupled plasma discharges relevant to the microelectronics industry. The models are trained on kinetic simulations over a range of device operating frequencies and pressures. Best performance was observed when simulations were initialized with ion kinetic profiles generated by a convolutional neural network, reducing the mean number of steps to reach convergence by $17.1\times$ when compared to initialization with a zero-dimensional global model. We also outline a workflow for continuous data-driven model improvement and simulation speedup, with the aim of generating sufficient data for full device digital twins.

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I. INTRODUCTION

Following a transient startup phase, many plasma systems operate in a quasi-steady, as opposed to a true steady state. Examples include systems driven by time-periodic forcing functions, those with persistent waves and instabilities, or chaotic and turbulent systems. Applications where such plasma states exist include electric propulsion devices,^{1,2} semiconductor manufacturing equipment,^{3,4} and fusion reactors which operate in a continuous (as opposed to a pulsed) mode.⁵ Further examples can be found across engineering, including fluid and structural mechanics,^{6,7} chemical,⁸ and electrical engineering.⁹ Due to the complexity of the equations involved, these systems are often modeled as initial-value problems, where the governing differential equations are numerically integrated until the system “converges” to a quasi-steady solution suitable for analysis. A major challenge associated with simulating plasma systems is the need to resolve both the fastest and slowest dynamics, necessitating a prohibitively large number of time steps to reach convergence. This multi-timescale challenge presents a significant bottleneck for computer-driven design of plasma

technologies, where rapid, cost-effective, and accurate exploration of parameter spaces is required.

The time to convergence of such systems is dependent on the chosen initial condition for the numerical integrator. As an example, consider a simple linear first-order ordinary differential equation,

$$x' + x = \cos t + \sin t + c, \quad (1)$$

where c is a fixed constant.

Assuming that some algorithm integrates this system starting from an initial condition $x(t=0) = x_0$. Then the continuous solution to the equation is,

$$x(t) = (x_0 - c)e^{-t} + \sin(t) + c, \quad (2)$$

a time periodic function with a decaying transient.

We seek the long-time quasi-steady solution: $x_\infty(t) = \lim_{t \rightarrow \infty} x(t) = \sin(t) + c$, which has time-averaged solution $\langle x_\infty(t) \rangle = c$. The error between the solution and the quasi-steady solution at any time is given as

$$\epsilon(t) = x(t) - x_\infty(t) = (x_0 - c)e^{-t}. \quad (3)$$

If the quasi-steady solution is sought with accuracy below some absolute tolerance $|\epsilon(t)| < \epsilon_c$, then Eq. (3) shows that the time to convergence $t_c \propto \ln |x_0 - c|$. Assuming that our numerical integrator adopts a uniform time step size, then it is clear that the closer the initial condition, x_0 , is to the time average of the quasi-steady solution c , the fewer steps will be required to reach convergence.

More generally, we could assume that $t_c \propto g(\mathbf{x}_0 - \mathbf{c})$, where $g(\mathbf{x})$ is a monotonic increasing function, and $\mathbf{x}_0$ and $\mathbf{c}$ denote the multi-dimensional initial condition and time-averaged quasi-steady solution, respectively. In a kinetic plasma system, the governing partial differential equations are either the Vlasov–Poisson or Vlasov–Maxwell systems, with appropriate collision terms and boundary conditions. Since these equations are tightly coupled and highly nonlinear, there exists no simple procedure for obtaining the optimal plasma simulation initial condition.

Data-driven approaches can offer a solution to this problem. More specifically, an initial condition generator can be trained on prior simulation data to select $\mathbf{x}_0 \approx \mathbf{c}$, thereby reducing the number of steps to convergence. For high-dimensional, multi-scale, or multi-physics problems, the amount of data required to learn an accurate mapping from design parameters to optimal initial conditions may be prohibitively large. This challenge can be mitigated by reducing the dimensionality of the initial condition, focusing on large-scale and long-time features of the system. Once a suitable low-dimensional initial condition is identified, the full system state can be recovered from the high-fidelity simulation, with accelerated convergence provided by the generated initial condition.

Data-driven Machine learning (ML) techniques are starting to see broader adoption within the plasma physics community¹⁰ including for low-temperature plasmas.^{11–17} The methodology described here is considered a type of hybrid machine-learning combined with traditional high-performance computing (ML+HPC),^{18–20} whereby the merits of both techniques are leveraged to accelerate simulations while retaining the same level of accuracy as first principles numerical methods. Examples of data-generated initial conditions for accelerating simulations include work by Hajgató *et al.*,²¹ who leveraged a convolutional neural network to realize a 13% speedup in fluid dynamics simulations for the optimization of aerodynamic design. Barros *et al.*²² relied on an automatic hyperparameter optimization tool to train a recurrent neural network, achieving an 8% speedup in solar wind forecasting models. Further examples of this relatively simple concept are sparse; however, we demonstrate that for highly multi-scale plasma systems, over an order of magnitude improvement in convergence time can be achieved.

While the outlined procedure has the ML component out-of-the-loop, there also exists a natural workflow for continuous model improvement that integrates ML more directly into parameter space exploration. A well-trained initial condition generator can be used to accelerate simulations, which can in turn be used to collect more data to improve the performance of the model. Iteration of this process may lead to sufficient data collection to produce a highly accurate reduced order model, or digital twin, of the system.²³ It should be emphasized that this approach is most practical when deep exploration of a parameter space is required, either for the purpose of design optimization or real-time control.^{24–26} As such, the procedure may prove

less beneficial for scientific discovery simulations when exploring completely new regions of a parameter space.

II. MOTIVATING EXAMPLE: CAPACITIVELY COUPLED PLASMA DISCHARGES FOR MICROELECTRONICS MANUFACTURING

To demonstrate the advantages of a data-driven initial condition generator, we consider applying the method to low-temperature plasmas used for the etching of silicon wafers. Plasmas are utilized in nearly 50% of the complex manufacturing steps required to turn raw silicon into microcircuitry,²⁷ including for the generation of extreme ultraviolet light for photolithography²⁸ as well as for chemical doping, cleaning, and etching of silicon surfaces.²⁹ Dry etching is commonly performed with Capacitively Coupled Plasma (CCP) discharges^{3,4} (see the schematic in Fig. 1). A time-periodic waveform between parallel plates (with frequency $\omega_{RF} = 2\pi F$) drives large-scale oscillations of plasma electrons, which are heated ohmically at high pressures, or stochastically at low pressures. Heated electrons then enable chemical reactions, which sustain the plasma. The heavier ions (with plasma frequency $\omega_{pi} \ll \omega_{RF}$) perceive the radio frequency (RF) sheath as a strong, constant electric field that accelerates them onto the silicon surface, etching intricate patterns into the substrate.

As traditional approaches for shrinking transistor size offer diminishing returns, the semiconductor industry is exploring more innovative approaches to sustaining Moore's law.³⁰ A prominent example includes complex three-dimensional microcircuit geometries, which can effectively stack circuit components vertically at the nanometer scale.³¹ High-aspect-ratio etching of deep channels into the wafer is an essential step to realizing these structures, requiring unprecedented control over ion kinetics at the plasma–surface interface.³² This has pushed the operating pressures of CCPs into the low or even sub-mTorr range,^{33–37} where improved ion velocity distribution function (IVDF) anisotropy can be achieved. Resolving these IVDFs, as well as electron stochastic heating and sheath-plasma instabilities³⁸ necessitates a kinetic treatment of the system. High-resolution first principles kinetic plasma simulations, therefore, play an essential role in understanding and optimizing CCP discharges to maximize wafer yield and throughput.

Beyond the requirement for high-dimensional kinetic resolution, high-fidelity simulations of CCPs are challenging due to the large mass ratio between electrons and ions, leading to the aforementioned

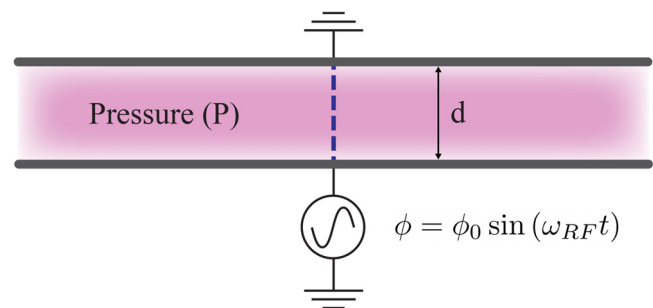


FIG. 1. Two-dimensional schematic of an argon capacitively coupled plasma discharge with gap width d , gas pressure P , driving voltage amplitude ϕ_0 and angular frequency $\omega_{RF} = 2\pi F$. The blue dashed line indicates the 1D spatial domain modeled by 1D-3V PIC simulations in this work.

difficulty of multi-timescale modeling. A requirement for numerical and physical accuracy is the resolution of the short timescale electron plasma period (ω_{pe}),^{38,39} whereas the system converges on the much longer timescale of ion transport. Kinetic simulations of CCPs are most commonly performed with the electrostatic particle-in-cell (PIC) algorithm augmented by Monte Carlo models for collisions between charged species and the neutral gas.^{39–43} The grid and particles in the PIC algorithm are highly amenable to parallelization and acceleration,^{44,45} however, the tightly coupled set of equations is advanced serially in time, limiting opportunities for parallelization and therefore further speedup. PIC simulations of CCPs are generally initialized with a uniform density Maxwellian distribution of electrons and ions, often informed by a zero-dimensional global model.^{3,4} This initial state is then relaxed to the quasi-steady solution via PIC iterations, usually requiring a minimum 10^6 and up to 10^8 time steps to reach convergence.

RF-CCPs often have the following ordering of time scales, $\omega_{pe} \gg \omega_{RF} \gg \omega_{pi}$, such that the perturbation of the IVDF about the mean solution is small and can be considered representative of a time-averaged steady-state solution for the CCP (the $\mathbf{c}$ from Sec. I). The goal is therefore to create an initial condition generator which can either predict the time-averaged IVDF (which we will simply refer to as f , without the species subscript) or moments of this distribution function. The implied hypothesis is that the lighter electrons will rapidly adapt to this initial state, and our simulation will converge in fewer time steps.

The paper proceeds as follows. Section III outlines the PIC code used to create the data and various machine-learning methods used to train the initial condition generator. The section also discusses how we measure convergence of our simulations and set a baseline for the proposed ML performance improvements. Section IV will then demonstrate that the proposed approach does indeed improve performance by over an order of magnitude. Section V will outline how this procedure can be integrated into a workflow for generating reduced-order models or digital twins of plasma reactors. Section VI concludes the findings of this research.

III. METHODOLOGY

A. Capacitively coupled plasma discharge simulations

To demonstrate the effectiveness of a data-driven initial condition generator for accelerating CCP simulations, we considered a reduced dimension, 1D-3V CCP configuration (see the dashed line in Fig. 1). Typical discharge parameters include the gas species, gas pressure (P), electrode gap (d), and, assuming a simple sinusoidal waveform, the voltage amplitude (ϕ_0) and frequency (F). For this demonstration, only F and P were allowed to vary with the species fixed to argon, $d = 5$ cm, and $\phi_0 = 300$ V. For each simulation, care was taken to ensure that the electron Debye length and plasma frequency were suitably resolved by the cell size and time step, respectively. Furthermore, the mean number of particles-per-cell for each species was above 300 for all simulations, providing sufficient statistical resolution.⁴⁶ We therefore assume that each simulation was numerically converged and the resulting profiles were independent of further numerical refinement, such that only physical parameters $\{F, P\}$ influenced the measured profiles. Unless specified, all simulations were initialized with a uniform Maxwellian distribution function with density n_0 , electron temperature T_{e0} , and ion temperature T_{i0} .

Simulations were completed using the 1D-3V Python PIC code `mini-pic`,⁴⁷ a parallel and accelerated code designed for prototyping

new algorithms. The code has been well benchmarked on previous CCP simulations^{47,48} and leveraged multiple CPU cores through `mpi4py`.⁴⁹ All simulations were run on the Princeton Stellar cluster, where each node consists of four 2.9 GHz Intel Cascade Lake CPUs, each with 24 cores.

The 210 simulations used for model training were nearly uniformly distributed across the $\{F, P\}$ parameter space with $\Delta P = 2.5$ mTorr and $\Delta F = \frac{1}{4} \times 13.56$ MHz (see additional details in Table I). A slightly finer resolution was used at lower frequencies and pressures close to the discharge ignition boundary. An additional 40 simulations used for testing of all initial condition generator models were randomly sampled across the parameter space. Simulations were run for a total time of $5000/F$, where convergence could be confidently guaranteed, and the final f profiles were computed by averaging the IVDF over 100 RF periods.

B. Definition for particle-in-cell simulation convergence

Due to the noisy and statistical nature of PIC simulations, defining a rigorous measure of simulation convergence is challenging. A useful decision algorithm for stopping a practical simulation should ideally be robust, cheap to compute, and performed online.

Attempting to satisfy these criterion, we consider the 1D steady state ion continuity equation,

$$\frac{\partial}{\partial x} \int v f dv \equiv \frac{\partial \Gamma_i}{\partial x} = S_+(x), \quad (4)$$

where $\Gamma_i(x)$ is the ion flux density and $S_+(x)$ represents the ionization source term. Notably, the kinetic behavior of the plasma is implicitly embedded within the source term, since $S_+(x)$ depends on the electron distribution function. As a result, the equation is capable of capturing kinetic effects without the need for expensive diagnostics of the full kinetic distribution functions.

Integrating both sides yields,

$$\Gamma_i(x) = \int_0^x S_+(x) dx + A \equiv \Lambda(x). \quad (5)$$

TABLE I. Physical and numerical parameters for the set of 1D-3V CCP simulations used to test the initial condition generator models.

Parameter	Value
Ion species	argon
Gas pressure range (P)	2–30 mTorr
Gap width (d)	5 cm
Driving frequency range (F)	6.78–50.85 MHz
Driving amplitude (ϕ_0)	300 V
Initial electron temperature (T_{e0})	1.5 eV
Initial ion temperature (T_{i0})	300 K
Number of simulations	250
Number of cells (N_x)	64–1408
Number of steps (N_t)	$3–31 \times 10^6$
Step size (Δt)	7.12–236 ps
Simulated time (T_s)	189–1688 μ s
Mean particles-per-cell	>300 per species

The integration constant A is selected such that the mean of $\Gamma_i(x)$ and $\Lambda(x)$ over the discharge gap is zero. To reduce the effects of numerical noise, Γ_i and Λ values are time-averaged over a fixed-size sliding window $T_a = a/F$, with $a = 5$ found to be a good compromise for reducing noise, while providing sufficient temporal granularity. At the end of each window, the data can be used to test for convergence and make a decision on stopping or continuing the simulation, or be stored for later (offline) analysis. For CCP settings $\{F = 27.12 \text{ MHz}, P = 20 \text{ mTorr}\}$, Figs. 2(a) and 2(b) show examples of time-averaged Γ_i and Λ profiles early in the simulation and after convergence.

At each time step, the level of convergence is measured via a symmetric normalized Euclidean norm of the difference between both sides of Eq. (5),

$$\epsilon(t) = \frac{\|\Gamma_i(x, t) - \Lambda(x, t)\|_2}{\frac{1}{2}\|\Gamma_i(x, t)\|_2 + \frac{1}{2}\|\Lambda(x, t)\|_2}. \quad (6)$$

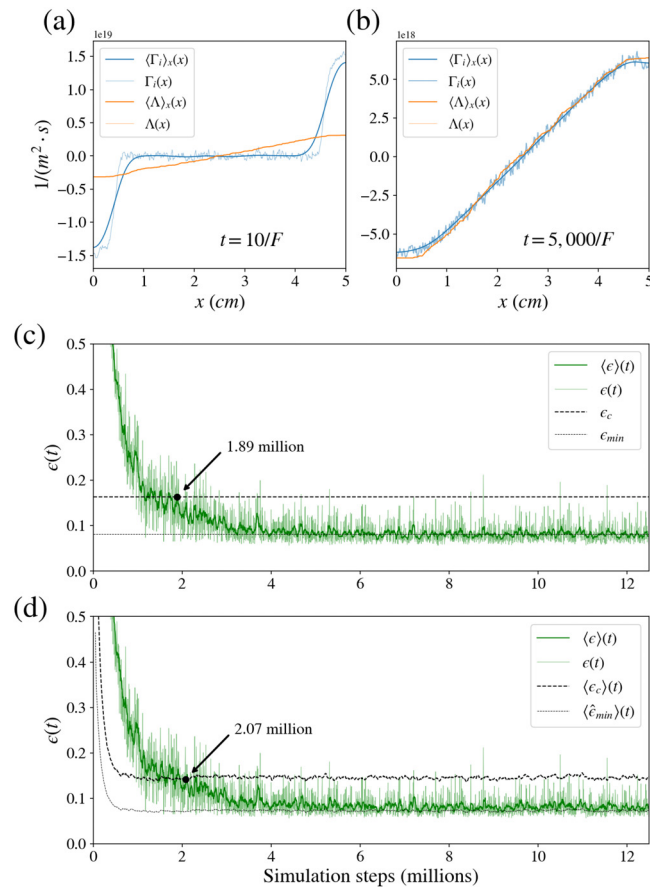


FIG. 2. (a) Plots of Γ_i and Λ early in the simulation (at 10 RF periods) and (b) after convergence has been achieved (at 5000 RF periods) for 1D-3V PIC simulations of an argon CCP operating at $F = 27.12 \text{ MHz}$ and $P = 20 \text{ mTorr}$. Thicker lines correspond to functions smoothed over space with a Gaussian filter. (c) Evolution of $\epsilon(t)$ with time, including the point of convergence measured by the offline procedure. The thicker line corresponds to the same data smoothed with a centered uniform filter. (d) Evolution of $\epsilon(t)$ and $\langle\epsilon\rangle(t)$ with time, including the point of convergence measured by the online procedure. Temporal smoothing is realized with a backward uniform filter.

Figure 2(c) shows a plot of $\epsilon(t)$ vs time for the same example simulation. Due to the noisy nature of PIC, $\epsilon(t)$ never truly converges to zero, but rather oscillates around some minimum value, which we will call ϵ_{min} .

1. Offline test for convergence

For an offline test of convergence, where the full temporal profile of error is known, ϵ_{min} is computed as the mean value of $\epsilon(t)$ for $t \in [\frac{3}{4}T_s, T_s]$. The level for convergence is set at two times this minimum noise level, such that $\epsilon_c = 2\epsilon_{min}$. To further reduce the effects of noise and outliers, $\epsilon(t)$ is smoothed via a centered uniform filter, which returns the mean across 11 time steps (denoted $\langle\epsilon\rangle(t)$). The convergence time is then measured as the last data point when $\langle\epsilon\rangle(t) \geq \epsilon_c$. Figure 2(c) shows the temporal evolution of these quantities and the point of convergence for the example simulation. Additional examples of convergence evolution for different CCP operating frequencies and pressures are provided in Appendix C.

2. Online test for convergence

An online test for convergence cannot rely on future information regarding $\epsilon(t)$, which complicates the procedure for measuring the minimum simulation noise level ϵ_{min} . To overcome this, we return to the spatial information contained within $\Gamma_i(x)$ and $\Lambda(x)$. First, spatially smoothed profiles of each function are obtained via a centered Gaussian filter with a standard deviation equal to $N_x/25$ (where N_x is the number of cells in the specific simulation). The underlying PIC noise minimum is then estimated as the sum of the norms of the residuals from each function, such that,

$$\hat{\epsilon}_{min}(t) = \frac{\|\Gamma_i(x, t) - \langle\Gamma_i\rangle_x(x, t)\|_2}{\|\Gamma_i(x, t)\|_2} + \frac{\|\Lambda(x, t) - \langle\Lambda\rangle_x(x, t)\|_2}{\|\Lambda(x, t)\|_2}, \quad (7)$$

where the estimate, $\hat{\epsilon}_{min}(t)$, is now a time-evolving quantity. The convergence limit is therefore also evolving in time as $\epsilon_c(t) = 2\hat{\epsilon}_{min}(t)$. To further reduce the effects of temporal noise, both $\epsilon(t)$ and $\epsilon_c(t)$ are temporally averaged over the preceding 10 periods (giving $\langle\epsilon\rangle(t)$ and $\langle\epsilon_c\rangle(t)$, respectively).

Another challenge associated with an online test for convergence is in deciding whether a dip in the convergence signal below the convergence criterion is permanent or temporary. To address this, we assume that if $\langle\epsilon\rangle(t) < \langle\epsilon_c\rangle(t)$ for 25 consecutive periods (in this case $25a = 125/F$) then convergence has been achieved. Figure 2(d) shows the temporal evolution of these quantities and the point of online convergence for the example simulation.

The parameters chosen for this online procedure were selected through testing of the offline and online procedures over the 250 CCP simulations completed for this work. While these parameters proved to be robust over the range of CCP input parameters, we recommend that care be taken when adopting this procedure, and tuning may be required for different problems and numerical settings.

C. Machine-learning techniques for initial condition generation

To reduce the dimensionality of our problem, we assume that the IVDF remains nearly isotropic in the directions parallel to the CCP electrodes, such that we consider only a 1D-1V $f(x, v)$ distribution (where $v \equiv v_x$). We also consider an even simpler model for predicting

the 0th moment of the IVDF (ion density) defined as $n(x) = \int f(x, v) dv$. The goal is to learn three mappings,

$$\Theta_1 : \{F, P\} \rightarrow n(x), \quad (8)$$

$$\Theta_{2,3} : \{F, P\} \rightarrow f(x, v). \quad (9)$$

For the lower dimensional Θ_1 mapping, we directly apply a classic Multilayer Perceptron (MLP) architecture, and for the second mapping, we consider two approaches; a Principal Component Analysis (PCA) decomposition of the data in combination with an MLP (Θ_2), as well as a Convolution Neural Network (CNN) architecture (Θ_3). Descriptions of each model are provided below with quantitative details in Table II.

Data were prepared by interpolating PIC particle densities onto a 1D–1V phase space grid, such that $f_{j,k} = f(x_j, v_k)$, where each x_j and v_k represent nodes of the grid. x_j spans the 5 cm gap with N_x points and v_k spans $-\frac{1}{2}v_{i,RF}$ to $\frac{1}{2}v_{i,RF}$ with 512 points, where $v_{i,RF} = \sqrt{2e\phi_0/m_i}$ is the velocity of an ion after acceleration through the RF potential amplitude. For the PCA+MLP model (Θ_2), the data were then mapped to a 400×512 grid, and for the CNN model (Θ_3) to a 128×128 grid. For the MLP model (Θ_1), $f(x_j, v_k)$ was numerically integrated in the v_k coordinate to obtain density profiles $n_j = n(x_j)$. Additionally, since ion density in CCPs exhibits a power-law dependence on input parameters F and P ,⁵⁰ the data were transformed into a logarithmic scale (using a softplus function), to improve learning performance, although this was not found to be necessary for the CNN model Θ_3 .

1. Training an MLP to predict ion density

Of the 250 simulations available, 180 were used for training, with 30 reserved for validation and hyperparameter tuning, and 40 for final testing. The MLP was implemented via the PyTorch framework,^{51,52}

incorporating three hidden fully connected layers, each with 64 neurons and ReLU activation functions. A Mean Squared Error (MSE) loss function combined with a softplus provided good performance and ensured positivity of generated density outputs. The Adam optimizer with a learning rate of 0.002 provided good convergence within 500 epochs. Full details of the network and optimized hyperparameters are provided in Table II. Training was extremely cheap, and could be completed on a laptop CPU in under 10 s.

2. Training a PCA and MLP model to predict IVDFs

To improve model accuracy, only simulations that produced non-zero final density were considered, with 195 used for training and 39 reserved for validation and testing. As shown in Fig. 4, the IVDF profiles have a central low-energy high-density region and two tails associated with ion acceleration through the RF sheaths. A PCA decomposition⁵³ was leveraged to reduce the dimensionality of the data for improved training with the MLP. PCA seeks a decomposition that exposes the maximum amount of variance across the dataset (i.e., across all simulations used for training) for a given number of components. The decomposition is computed on the training data only, and whitening was used to better normalize each component. A threshold of 91% explained variance was used, requiring 27 PCA components.

The respective weights of each PCA component were then used to train the MLP, with the architecture and optimal parameters shown in Table II. Generation of IVDF profiles was achieved by outputting PCA weights from the trained MLP. The full reconstruction of the IVDF was then obtained by expanding the PCA components with their respective weights. A workflow for the data curation and model inference procedures is outlined in Fig. 3(a). Total training time was

TABLE II. Data and hyperparameters used to train the ML-based initial condition generators.

Model	Θ_1 - MLP	Θ_2 - PCA+MLP	Θ_3 - CNN
No. train samples	180 (72%)	195 (83%)	195 (83%)
No. validation/test samples	30/40 (28%)	39 (17%)	39 (17%)
Training data type	n_j $j \in [1, 16]$	$f_{j,k}$ $j \in [1, 400]$ $k \in [1, 512]$	$f_{j,k}$ $j \in [1, 128]$ $k \in [1, 128]$
PCA explained variance	...	91%	...
PCA components	...	27	...
Optimizer	Adam	Adam	Adam
Loss function	MSE & softplus	MSE	MSE
Activation function	ReLU	ReLU	CELU
No. epochs	500	3000	1000
Learning rate	2×10^{-3}	2.2×10^{-4}	1×10^{-3}
Input parameters	$\{F, P\}$	$\{F, P\}$	$\{F, P\}$
No. MLP hidden layers	3	4	6
MLP hidden layer widths	[64, 64, 64]	[64, 128, 256, 128]	[10, 64, 256, 1024, 2048, 2048]
CNN layer kernel size	...	...	4
CNN layer stride	...	...	2
CNN layer padding	...	...	0
CNN layer output channels	...	...	[16, 8, 4, 1]
Output data type	Same as training	Same as training	Same as training

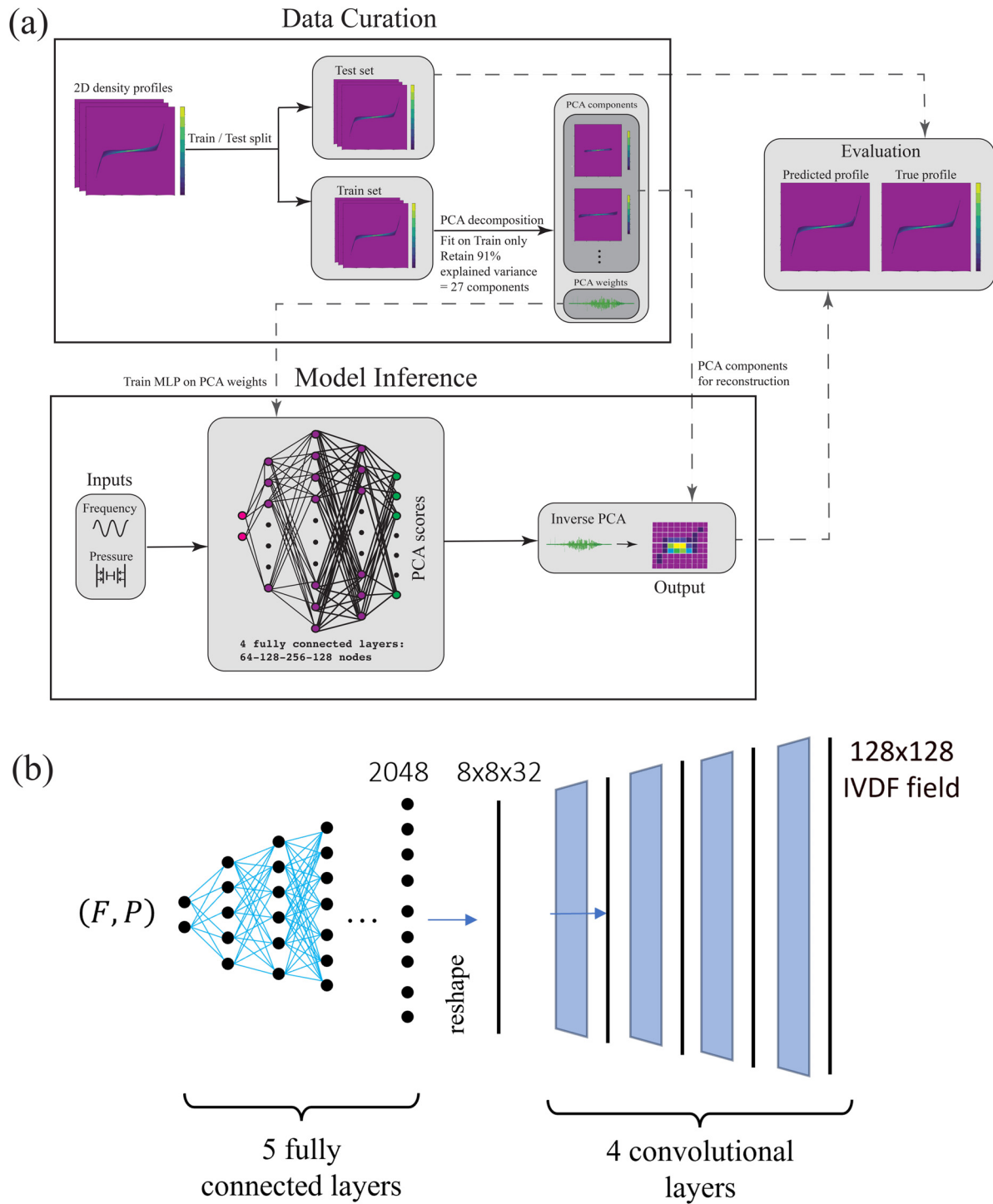


FIG. 3. (a) Workflow and model structure for the PCA+MLP data curation and inference of IVDF profiles. The data are initially split into train and test samples. Dimensionality is reduced via a 27-component PCA decomposition, retaining 91% of explained variance. The weights of these components for a given CCP frequency and pressure are then learned via an MLP with four hidden layers. Initial conditions are generated by first predicting the PCA weights from the MLP and then reconstructing the profiles from the PCA components. (b) Model structure of the CNN trained to produce initial conditions for the IVDF. The model includes six fully connected layers, followed by reshaping, and then four convolutional layers.

90 s on a single NVIDIA A100 GPU, a speedup of $48\times$ compared to training on a single core of a 2.9 GHz Intel Cascade Lake CPU.

3. Training a CNN model to predict IVDFs

Similarly, 195 simulations (with non-zero density) were used for training, and 39 were reserved for validation and testing of this model. A convolutional neural network leverages the structure of the data in a different way than a PCA decomposition. The model directly includes layers that “learn” the spatial structure of the data through convolutional or deconvolutional operations (in practice, interpolation). Although the data here is two-dimensional, the convolutional operations include a third-dimension, known as “channels,” which learn and capture different features of the data. Figure 3(b) shows the structure of the CNN model, which first includes a 6-layered MLP to upsample the CCP input parameters before reshaping the data for the deconvolutional layers. Full details of the model are provided in Table II. Total training time was 20 s on a single NVIDIA A100 GPU.

D. Establishing a baseline and measuring simulation speedup

Establishing an appropriate baseline for machine-learning-enabled improvements of simulation performance is an essential part of evaluating the merits of any ML technique.⁵⁴ Since we consider approaches to improve the initialization of CCP simulations, this amounts to making an assumption on how PIC simulations for CCP discharges are typically initialized. In the authors’ experience, most modelers will initialize simulations with a uniform density Maxwellian distribution of electrons and ions, similar to those used for data collection in this paper. Choosing a practical initial density n_0 and species temperatures T_{e0} , T_{i0} can be determined from the literature or the modelers previous experience with similar simulations. To define a more rigorous baseline, we rely on a 0D non-uniform CCP global model³ to predict n_0 and T_{e0} , with ion temperature always initialized at 300 K. Full details of the global model and solution algorithm are provided in Appendix B.

The potential speedup provided by each of the initial condition generators when compared to this baseline was measured on the reserved test data that was not used during training. Each of these CCP simulations was run with six different initial conditions,

- (1) A uniform density determined by the 0D global model.
- (2) The exact time-averaged density profile obtained from converged simulations.
- (3) The exact time-averaged IVDF profile obtained from converged simulations.
- (4) The density profile (with Maxwellian velocity distribution) determined by Θ_1 (MLP).
- (5) The IVDF profile determined by Θ_2 (PCA+MLP) and the electron density profile determined by integrating the IVDF.
- (6) The IVDF profile determined by Θ_3 (CNN) and the electron density profile determined by integrating the IVDF.

PIC initial conditions were created via rejection sampling over the respective n or f profiles.

At a single operating condition, simulation speedup (or slow-down) was measured as the ratio of the number of steps required to reach convergence (either by the online or offline procedures outlined

in Sec. III B). Speedup for each simulation was always measured against the baseline, that is, the number of steps required for a simulation to reach convergence when initialized with a uniform density predicted by the global model. Across the entire dataset, speedup was measured as the mean speedup for each model type.

We stress that the final converged state of each CCP simulation is identical, within numerical tolerance, regardless of the initial condition chosen. We therefore consider the described approach to be a strong baseline for measuring ML model performance.

IV. RESULTS: ACCELERATING SIMULATION CONVERGENCE

A. Convergence time with ideal initial conditions

To verify that improved initial conditions can accelerate kinetic RF-CCP simulations, we begin by considering a single point in the CCP parameter space $\{F = 27.12 \text{ MHz}, P = 20 \text{ mTorr}\}$. Figure 4(a) shows the resulting time-averaged (over 100 RF periods) steady-state ion density profile n (blue line), and Fig. 4(b) the time-averaged steady-state IVDF f .

The simulations are run to convergence with various initial conditions,

- (1) A series of uniform density profiles, as shown by the dashed lines in Fig. 4(a).
- (2) The exact time-averaged ion density profile, as shown by the solid blue line in Fig. 4(a).
- (3) The exact time-averaged IVDF, as shown in Fig. 4(b).

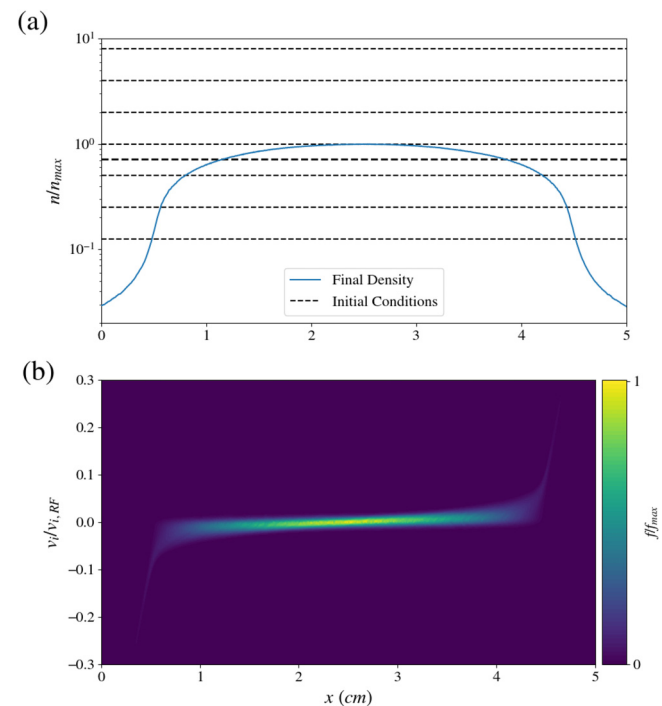


FIG. 4. (a) Converged time-averaged ion density profile (blue line) of an argon CCP discharge operating at 27.12 MHz with 20 mTorr gas pressure and a range of uniform initial conditions (dashed lines). The thicker dashed line corresponds to the density predicted by the global model. (b) Converged time-averaged ion velocity distribution function (IVDF) for the same discharge parameters.

Electrons are initialized with the same density profile as ions, either directly from n or by integrating f , and with an isotropic Maxwellian velocity distribution. Unspecified components of the IVDF are also initialized in the same way (i.e., in the v_y, v_z directions) with Maxwellian distributions using the characteristic temperatures given in Table I.

Figure 5(a) shows the evolution of the convergence error $\epsilon(t)$, as defined by Eq. (6), vs time step for all initial conditions. The convergence time data are summarized in Fig. 5(b) with the x axis showing the ratio of initial density to the peak ion density of the time-averaged profile n_{max} . The steps to convergence for the exact ion density and velocity distribution function initial conditions are placed under the $n_0/n_{max} = n_{gm}$ point, which is the density predicted by the global model.

Figure 5 demonstrates that, for this single point in the parameter space, using the one-dimensional ion density data as an initial condition offers a $6.8\times$ speedup over even the best choice in uniform initial condition (which occurs when $n_0/n_{max} = n_{gm}$). This indicates that higher-order moments of the ion distribution function, as well as the quasi-steady electron distribution function, reach convergence much faster than the ion density profile. Initializing a simulation with the two-dimensional IVDF data offers even better performance with a $27\times$ speedup in convergence. These results demonstrate that improved initial conditions can result in over an order of magnitude reduction in the number of time steps required to reach convergence.

The figure also reveals several other noteworthy trends. In particular, when initializing with a uniform ion density, the fastest convergence is achieved when the initial value matches that predicted by the global

model. Overestimating the initial density, i.e., choosing a value higher than this optimal value, quickly degrades convergence performance. The trend is approximately logarithmic, similar to that from the example Eqs. (1)–(3) presented in Sec. I. Despite this supposedly weak, logarithmic dependence on initial condition, we note that the number of steps required to reach convergence is still on the order of 10s of millions. From a physical standpoint, the observed trend is likely due to the fact that excess ions must be transported out of the system to achieve the correct final profile. However, selecting an initial uniform density below the optimal value results in only a modest increase in convergence time. In this case, the system generates new ions near where additional density is required, minimizing the need for long-range ion transport. The takeaway is that when employing a uniform density initial condition, a simple global model estimate might offer the best performance (see Appendix B for implementation details), and it is generally preferable to underestimate rather than overestimate this value.

The insert in Fig. 5(a) also shows that the simulation initialized with the true f reaches convergence within the first tested time step (i.e., the width of the temporal averaging window defined in Sec. III B 1). This suggests that a convergence test with higher temporal resolution may measure an even greater speedup in convergence time.

B. Convergence time with ML-generated initial conditions for ion density

With this validation of the underlying concept for our approach, we proceed to test the MLP model described in Sec. III C 1 as an initial

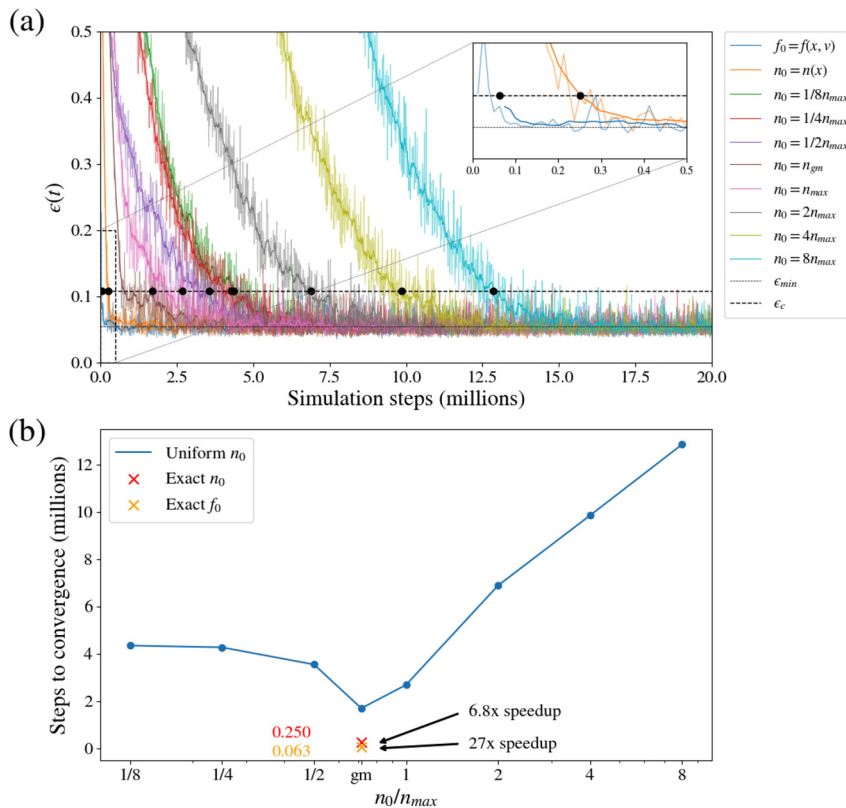


FIG. 5. (a) Convergence error vs number of simulation time steps for various initial conditions. Black dots indicate the point where convergence is achieved for each simulation with respect to the offline test procedure. (b) Steps to convergence for various initial conditions vs the ratio of initial density to the peak density of the steady-state. The red and orange crosses indicate the number of steps required when initializing the simulation with the exact final ion density profile n and final IVDF f , respectively. The stated speedup values for these crosses are with respect to the blue dot above at $n_0/n_{max} = n_{gm}$.

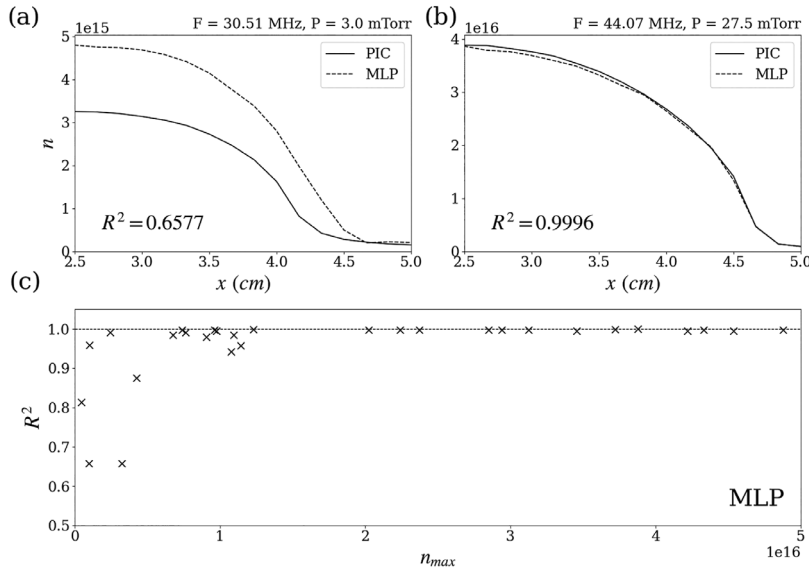


FIG. 6. PIC simulation vs MLP-generated time-averaged ion density profiles (on the half domain) for (a) the worst R^2 value and (b) the best R^2 value, and (c) plots of R^2 for the 40 test cases vs peak profile density (n_{max}) indicating that the MLP performs worst at generating low-density profiles.

condition generator for n profiles. Figure 6 shows several half n profiles from the test set, including their true curves and those predicted by the MLP. These include the worst fit [Fig. 6(a)] as well as the best fit [Fig. 6(b)], which is more typical of most results. Figure 6(c) plots the R^2 error of each test case against peak density, revealing that the model performed worst when predicting lower density (lower n_{max}) profiles. Possible explanations for this behavior include that the absolute MSE, rather than the normalized MSE, was chosen for the loss function. Additionally, at low pressures and frequencies, near the cutoff for discharge ignition, the sheath becomes larger, resulting in narrower density profiles, which were represented by very few simulations in the training data. This deficiency could be ameliorated by training on additional simulations near the cutoff boundary.

Fortunately, simulations with lower density are both computationally cheaper and converge more rapidly, even when using typical uniform initial conditions. Therefore, the model falls short within a parameter range where its purported improvements are least needed. Across the entire test set, the model achieved a mean R^2 score of 0.958 and median of 0.996, demonstrating its ability to accurately generate n profiles.

When the 40 test cases were initialized with MLP-generated initial conditions they on average converged, using an offline metric, $5.1 \times$ faster than those initialized with a uniform density predicted from the global model. When using an online test for convergence, this value decreased to $2.5 \times$. To determine the maximum theoretical speedup possible for this approach, each simulation was also initialized with its exact time-averaged quasi-steady-state ion density profile, providing a mean speedup of $6.5 \times$ ($2.9 \times$ online). Therefore, the MLP-driven ML model for predicting ion density initial conditions achieved 78% (88% online) of the maximum possible reduction in simulation steps.

C. Convergence time with ML-generated initial conditions for IVDF

Turning to predictions of the 1D-1V f we consider the performance of the PCA+MLP and CNN architectures, as described in

Secs. III C 2 and III C 3, respectively. Figure 7 shows that both models produce similar trends in R^2 error, with the worst performing predictions occurring for the lowest maximum phase space density f_{max} . This is likely due to the same issues with an MSE loss function and poor sampling near the discharge cutoff region, as observed for the ion density model. The mean R^2 error for the PCA+MLP model was 0.660; however, the median was 0.983, which can be explained by the outliers observed in the figure at very low phase space densities. The CNN model had a mean and a median R^2 errors of 0.991 and 0.998, respectively, indicating that this model had overall improved predictive capabilities. Both models provided good predictions for most cases, and we refer the reader to Appendix D for select visual comparisons between predicted and true f profiles.

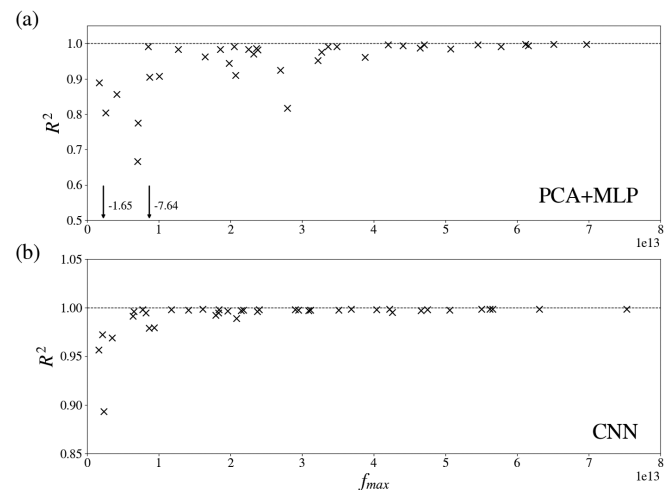


FIG. 7. R^2 vs peak phase space density (f_{max}) for the 39 non-zero test cases for (a) the PCA+MLP model and (b) the CNN model. The position of outliers in (a) is denoted by the arrows and associated numerical values.

On average, test cases initialized with the PCA+MLP initial condition generator converged, using an offline metric, $7.6\times$ faster than those with a uniform density initial condition predicted by the global model. When using an online test for convergence, this value decreased to $2.9\times$. For the CNN model, convergence was achieved $17.1\times$ faster using the offline test, whereas using an online test, the speedup was $4.4\times$. To determine the maximum theoretical speedup possible with this approach, each simulation was also initialized with the exact time-averaged steady-state IVDF, providing a mean speedup of $19.5\times$ ($4.6\times$ for online convergence). Therefore, the PCA+MLP model achieved 39% (63%) of the maximum possible reduction in simulation steps, and the CNN model achieved 88% (96% online) of the maximum possible reduction in simulation steps.

D. Summary and discussion of speedup provided by ML-generated initial conditions

Table III summarizes the model accuracy and mean simulation speedup achieved by each of the simulation initialization approaches. A broad conclusion that can be drawn from these results is that machine-learning generated initial conditions can accelerate the time to convergence for kinetic simulations of capacitively coupled plasma discharges. Peak speedup was found to occur for simulations initialized with CNN model-generated IVDFs, providing $17.1\times$ speedup when measured with the offline convergence criterion and $4.4\times$ with the online convergence criterion.

Generated profiles of ion density n using MLPs can achieve near peak speedup when compared to initialization with the exact profiles. With respect to the 2D IVDF initial conditions, we see different performance levels between the two models. The CNN model performs best overall, reaching near peak speedup when compared to the exact IVDF profiles. The PCA+MLP shows reduced performance when compared to the CNN model; however, it still outperforms the MLP model for density alone. While the PCA+MLP model performed well at estimating most initial conditions, the outlier cases [see Fig. 7(a)], whose generated IVDF profiles led to longer convergence times, skewing the mean performance numbers. Improvements may be possible by increasing the threshold of explained variance, resulting in additional components in the PCA decomposition.

The results also clearly demonstrate that realizing these performance improvements in practice will be limited by the technique chosen for online testing of simulation convergence. Compared to the offline convergence criterion, the online approach leaves approximately $3\times$ of the available performance improvement on the table.

The largest differences in online and offline speedup occur for the IVDF initial conditions, primarily due to the very fast convergence times of simulations initialized with the PCA+MLP, and particularly, the CNN-generated initial conditions. As shown within the insert axes of Fig. 5(a), simulations can often converge quicker than the temporal averaging period used to check convergence, making the measured convergence time highly sensitive to the choice of averaging period, as well as the chosen waiting time to ensure that the error does not bounce back over the designated threshold.

To provide some qualitative examples of how the convergence criterion can influence measured speedup time, we consider altering some of the parameters set for offline and online convergence, as defined in Sec. III B. First, considering the offline convergence criterion, when reducing the averaging window for $\epsilon(t)$ from 10 to 5 periods, and increasing the convergence threshold from $2\epsilon_{min}$ to $3\epsilon_{min}$, the CNN initial condition generator provides a $26\times$ speedup over uniform initial conditions when compared against the global model (compared to $17.1\times$ reported in Table III). Further decreasing the averaging window to just 1 period, and the convergence threshold to $4\epsilon_{min}$, provides a measured speedup of $38\times$. Turning to the online convergence criterion, reducing the backward averaging window of $\epsilon(t)$ from 10 to 5 periods, the convergence threshold from $2\hat{\epsilon}_{min}(t)$ to $3\hat{\epsilon}_{min}(t)$, and reducing the number of consecutive periods required for convergence from 25 to 10, we see a speedup of $6.4\times$ (compared to $4.4\times$ reported in Table III). Further reducing the averaging window to 1 period, and the required number of consecutive periods under the threshold to 5 (while leaving error levels unchanged), provides a measured speedup of $8.8\times$. Of course, choosing more relaxed definitions of convergence reduces the modeler's certainty as to when convergence has actually been achieved, the user should choose their criterion based on the given application and accuracy requirements. We further emphasize that future research should be directed toward more efficient techniques for measuring PIC simulation convergence.

The machine-learning techniques used in this work have proven to be effective at generating accurate initial conditions for PIC simulations, with the CNN model achieving 88% and 96% of maximum speedup for the offline and online convergence criterion, respectively. When considering sparser datasets or larger parameter spaces, however, more advanced techniques may prove advantageous. Notable techniques include physics informed neural-networks, whereby physical laws or constraints are incorporated into the loss function,^{55,56} U-Nets, which offer improved multiscale representation,^{57–59} and diffusion models,⁶⁰ which have become the de facto technique for image generation (analogous to the n and f profiles considered here).

TABLE III. Summary of initial condition generator performance for accelerating kinetic simulations of CCPs.

Initial Condition	No. sims training	No. sims testing	Mean, R^2	Median, R^2	Offline speedup	Online speedup
Uniform GM	...	40	...	...	1.0	1.0
Exact n	...	40	...	...	6.1	2.9
MLP for n	210	40	0.958	0.996	5.1	2.5
Exact f	...	39	...	...	19.5	4.6
PCA+MLP for f	195	39	0.660	0.983	7.6	2.9
CNN for f	195	39	0.991	0.998	17.1	4.4

Regarding the physics of a CCP discharge, a number of complicating factors may require additional data collection or the use of more advanced ML approaches mentioned above. This includes the case of multiple ion species, such as within industry relevant CF_4 plasmas, which would require learning of mappings between input parameters and IVDF profiles for each species. Some CCP discharges also exhibit hysteresis, where different initial conditions can result in different converged solutions. For example, when two different steady-states are observed,⁶¹ training of an ML classifier may be required to first determine the operating regime of the discharge based on the initial condition, followed by application of similar regression algorithms to those explored in this work. Finally, there is the particularly challenging case of when the driving frequency becomes comparable to the ion plasma frequency. In this case, the IVDF will evolve periodically (similar to electrons), diminishing the usefulness of initializing a simulation from a single steady state profile. Under such circumstances, it may prove necessary to learn the IVDF at a fixed phase of the RF cycle. This would also require learning of the EEDF to ensure that the resulting initial electrostatic field is consistent with the plasma state. The challenge with this approach would be in collecting sufficient data from each simulation to produce a statistically well-resolved distribution, since only 1 time step of each RF period can be used to produce a smooth distribution from the noisy PIC data.

V. WORKFLOW FOR DIGITAL TWIN MODEL GENERATION

At this point, it would not be unreasonable for the reader to raise questions regarding the overall computational efficiency of the approach outlined thus far. Running hundreds of simulations for the purpose of accelerating a few additional simulations is clearly a poor use of computational resources. However, we reiterate that the purpose of this methodology is to accelerate convergence of costly first principles simulations for computer-aided engineering of complex devices, with the example here being a CCP discharge for silicon dry etching. The approach will prove computationally advantageous when a large number of simulations are required to explore a relatively well-defined range of input parameters.

We also reemphasize that the ultimate goal of these simulations is not necessarily to obtain an ML model for the relatively low-order behavior embedded in the initial conditions, but rather to accelerate the solution of the more complex physics obtained within the high-fidelity simulations. For silicon etching, this is the precise ion flux, energy, and angular distributions at the interface between the plasma and the wafer, as well as the electron energy distribution functions which drive chemistry within the bulk plasma. The relationship between the input parameters and these quantities are complex and highly nonlinear, and would require significantly more data and/or more sophisticated techniques to generate a reduced order model or digital twin which can capture these effects.

Figure 8 demonstrates a workflow by which traditional first-principles physics simulations (designated as “Sims” in the figure) could be used in combination with ML techniques to accelerate the collection of data for the construction of a reduced-order model or digital twin. The modeler first identifies an appropriate parameter space over which to train the model. In this paper, that is a range of frequencies F and gas pressures P as inputs for a CCP discharge. The workflow begins by first exploring this parameter space using sparsely sampled PIC simulations. Data from these simulations are then used to train a

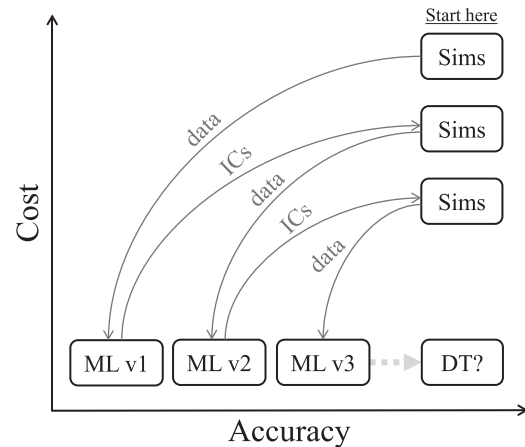


FIG. 8. Workflow for reducing the cost of first principles simulations (designated as “Sims”), while simultaneously improving the accuracy of an ML model for initial condition generation. Repetition of this cycle could help produce a reduced-order model, or digital twin (DT), with sufficient accuracy to enable design optimization or real-time control.

first version of the ML model, which in turn is used to accelerate the next round of PIC simulations applied to explore the parameter space with greater resolution. Iterating on this methodology would allow for progressively faster PIC simulations (shown as the Sim box moving down the “Cost”-axis in Fig. 8) and improved ML model accuracy (shown as the ML box moving right on the “Accuracy”-axis). Eventually, with sufficient data collection, the ML model may prove suitably accurate to avoid running PIC simulations for many regions of the parameter space. Such a model is represented, aspirationally, by the “DT” (Digital Twin) box shown in the figure. This model could then be called on to rapidly prototype within the explored parameter space, or enable real-time control and optimization of device performance. The high-fidelity PIC simulations would remain as a backup when additional resolution or certainty is required.

An implied hypothesis in this procedure is that the ML model will enable faster simulations, even when the parameter space is sparsely sampled. To show that this is true for CCP simulations accelerated by initial conditions, the CNN model described in Sec. III C 3 was trained on increasingly sparse datasets, and tested for its ability to accelerate convergence of the 39 (non-zero) test simulations, the results of which are shown in Table IV. It was observed that even with an extremely sparse sampling of 3×3 samples across the parameter space, the model could realize a $2.7\times$ speedup with offline convergence and $2.0\times$ speedup with online convergence when compared to initialization with uniform density estimated by the global model. Moreover, speedup improvements continue until saturation at the levels reported in Sec. IV C. Estimating the compounding effects of this workflow, as outlined in Fig. 8, suggest that collecting data for the entire set of 195 (non-zero) simulations could be accelerated by a factor of $3.6\times$, assuming online convergence testing, or $8.2\times$ if a convergence test approaching the offline speedup measures could be implemented. Table IV also reveals that our model saturates in performance after incorporating data from only 56 simulations, indicating that fewer runs may have been required to realize the speedups reported in Sec. IV.

TABLE IV. CNN accuracy and PIC simulation speedup achieved at various iterations of the digital twin training workflow shown in Fig. 8. In this instance, the model architecture remains fixed, and each version of the model simply incorporates additional data.

Iteration	No. sims training	Mean, R^2	Median, R^2	Offline speedup	Online speedup
I	9	0.802	0.942	2.7	2.0
II	16	0.940	0.992	7.2	3.4
III	25	0.925	0.995	10.5	3.9
IV	42	0.943	0.996	12.8	4.0
V	56	0.967	0.997	16.9	4.4
VI	195	0.991	0.998	17.1	4.4

There are some important challenges to consider when deploying this procedure to a larger or higher-dimensional parameter space. For one, the hypothesis regarding monotonic improvements in ML performance may not hold true for very sparse samples over a large parameter space, potentially requiring a larger upfront cost for initial, non-accelerated, simulations prior to beginning the procedure outlined in Fig. 8. Alternatively, the version numbers associated with each ML model in the figure need not only refer to model improvement associated with training on additional data, but could also refer to different model architectures or machine learning techniques entirely (see Sec. IV D for further discussion). The ML model chosen for each level should be optimized to take advantage of the amount of available data.

Another consideration is how best to sample from a larger or higher dimensional parameter space. Simulations in this work were sampled nearly uniformly across the two-dimensional space, an approach which could prove sub-optimal in higher dimensions. Random sampling, Latin hypercube sampling^{62,63} or space filling curves⁶⁴ offer viable approaches. Furthermore, if the ML method can be modified to provide uncertainty quantification, Bayesian optimization⁶⁵ techniques may select sample locations which best improve model performance. A suitable stopping criterion for data collection could be met once PIC simulation speedup has saturated, or when an ML model for the desired output (such as surface IVDFs) has reached a sufficient accuracy. Exploring these ideas will be the subject of future research.

VI. CONCLUSION

This paper has demonstrated that machine learning techniques can be applied to develop an accurate mapping between the operating parameter space of capacitively coupled plasma discharges and the final time-averaged ion density and velocity distribution profiles. Initializing further kinetic simulations with generated profiles resulted in a mean speedup of up to 17.1× when compared to initialization with a uniform-density global model. The best performing model was a convolutional neural network for generating ion kinetic profiles with a mean R^2 of 0.991. Accelerating first principles simulations via this method will prove most useful when deep exploration of a given parameter space is required, as is common for computer-aided engineering design. By reducing the computational cost associated with collecting further simulation data, a natural workflow was demonstrated that could assist in the development of robust reduced order models, or eventually, digital twins which could be used for online optimization and control.

This work offers many opportunities for extension, including exploration of a wider parameter range, learning of higher fidelity

initial conditions (perhaps with temporal characteristics of the distribution functions), and applying the 1D(-1V) results to generate initial conditions for more costly higher dimensional simulations. The procedures demonstrated here can also likely be applied to other multi-timescale systems, providing improvements in simulation performance and reduced time for design of new technologies.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Andrew T. Powis: Conceptualization (lead); Data curation (lead); Formal analysis (equal); Investigation (equal); Methodology (equal); Software (lead); Validation (lead); Visualization (equal); Writing – original draft (lead). **Domenica Rivera:** Conceptualization (supporting); Formal analysis (equal); Investigation (equal); Methodology (equal); Software (supporting); Visualization (equal); Writing – original draft (supporting). **Alexander Khrabry:** Conceptualization (supporting); Formal analysis (equal); Investigation (equal); Methodology (equal); Software (supporting); Visualization (equal); Writing – original draft (supporting). **Igor D. Kaganovich:** Conceptualization (supporting); Formal analysis (supporting); Funding acquisition (lead); Project administration (lead); Resources (lead); Supervision (lead); Writing – original draft (supporting).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

APPENDIX A: WHY GPUS ARE A BOTTLENECK FOR FASTER TIME-STEPS

Although the grid and particles within a PIC simulation are well-suited for parallelization and hardware acceleration, the tightly coupled nature of the equations means that time-stepping is inherently serial. If each time step of the simulation takes up a fixed computer wall-clock-time, then this sets a lower bound on the total simulation runtime. In this paper, we explored the use of an initial condition generator to reduce the number of time steps to reach convergence, however the reader could just as well ask. Why don't we reduce the wall-clock-time per step? Interestingly, the increasing reliance on modern accelerators such as GPUs tends to constrain the potential gains available through this approach.

When adapting an algorithm for execution on a GPU, the prevailing best practice is to expose as much parallelism and data throughput as possible, thereby leveraging the chip's massively parallel architecture. For suitably structured algorithms, this yields significant performance improvements, even though individual GPU cores operate at lower clock speeds compared to typical CPU cores. However, this architectural design comes with limitations, specifically, diminished strong scaling efficiency relative to CPUs [see Fig. 9(a)], which in turn restricts the performance benefits achievable through increased resource allocation. Nonetheless, as illustrated in Fig. 9(b), GPUs consistently outperform CPUs across all node counts, demonstrating a clear net speedup despite their scaling constraints.

The intent of this discussion is to highlight the limitations users face in further accelerating individual time steps on modern HPC architectures. While GPU-based nodes remain the most effective option for minimizing total wall-clock time (not to mention energy usage), their limited strong scaling capabilities restrict the extent to which performance can be improved through additional resource allocation. This limitation motivated the approach pursued in this work, which rather aims to reduce the number of time steps required to reach a desired level of accuracy or convergence.

APPENDIX B: INHOMOGENEOUS ARGON CAPACITIVELY COUPLED PLASMA GLOBAL MODEL

The global model used to predict plasma density in Sec. IV is an implementation of the inhomogeneous global model for argon CCPs from Chabert³ Chap. 5, which draws directly from the work of Godyak⁶⁷ and Lieberman and Lichtenberg.⁴ The model is derived under the following assumptions:

1. The electron temperature T_e (given in units of eV) is constant in space.
2. Ion inertia is sufficiently large such that ions respond only to the time-averaged electric field.
3. Electron inertia is negligible. Electrons, therefore, follow the instantaneous electric field.
4. The system is electrostatic.
5. Gas pressure is constant in space and within the low to intermediate range (under 100 mTorr).

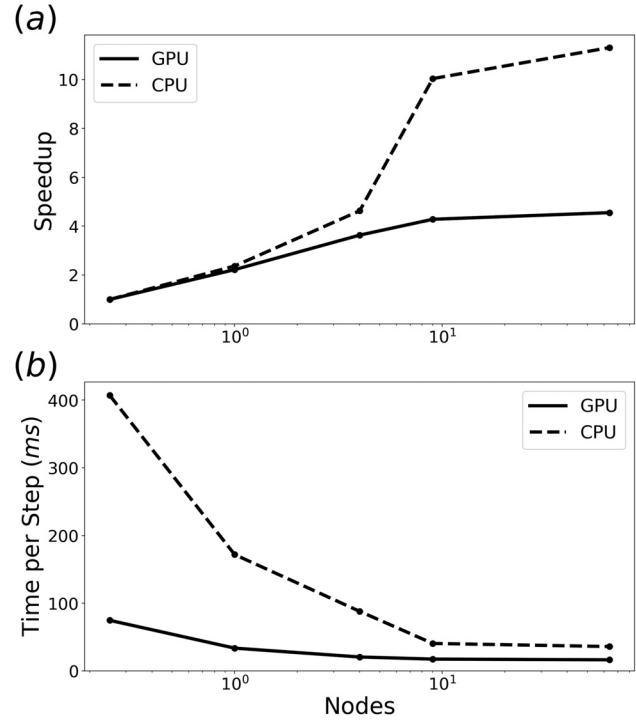


FIG. 9. Strong scaling performance of the LTP-PIC code⁶⁶ when modeling a two-species thermal plasma with all-periodic boundary conditions. The grid size is 256×256 and there are approximately 130×10^6 simulation particles. (a) Speedup plot demonstrating that CPUs offer improved strong scaling over GPUs, and (b) plot of time per step demonstrating that, despite scaling limitation GPUs still offer improved performance. Each node was comprised of $1 \times$ AMD EPYC 7763 CPU with 64 cores and $4 \times$ NVIDIA A100 GPUs. The leftmost data point was obtained using 1/4 of a node (16 CPU cores or 1 GPU).

6. There is no volume recombination or multi-step ionization.
7. The sheath size is small compared to the discharge length.

1. Particle balance

At lower pressures, the steady-state particle balance can be written as

$$n_g K_{iz}(l - s_m) = 2h_l u_B, \quad (\text{B1})$$

where n_g is the gas density, l is the discharge length, s_m is the sheath length and K_{iz} is the ionization rate coefficient defined by

$$K_{iz} = K_{iz0} \exp\left(-\frac{\varepsilon_{iz}}{T_e}\right), \quad (\text{B2})$$

where $K_{iz0} = 5.0 \times 10^{-14} \text{ m}^3/\text{s}$ and $\varepsilon_{iz} = 17.44 \text{ eV}$ is the argon ionization potential. h_l is the edge to center density ratio, which in the low to intermediate pressure regime is given by

$$h_l \approx 0.86 \left[3 + \frac{l}{2\lambda_i} \right]^{-1/2}, \quad (\text{B3})$$

where λ_i is the ion-neutral mean-free-path, which for argon can be approximated as

$$\lambda_i = \frac{31.5 \times 10^{-3}}{P}, \quad (\text{B4})$$

where P is the gas pressure in units of mTorr. u_B is the Bohm speed,

$$u_B = \left(\frac{eT_e}{M} \right)^{1/2}, \quad (\text{B5})$$

where M is the ion mass.

When solving the global model, we consider the particle balance equation in the following form:

$$\begin{aligned} f(T_e, s_m) &= \frac{n_g(l - s_m)}{2h_l} - \frac{u_B}{K_{iz}} \\ &= \frac{n_g K_{iz0}(l - s_m)}{2h_l} - \left(\frac{eT_e}{M} \right)^{1/2} \exp\left(\frac{\varepsilon_{iz}}{T_e}\right). \end{aligned} \quad (\text{B6})$$

2. Power balance

The steady-state electron power balance can be written via a circuit representation. Since our system is voltage driven, we consider the power balance in terms of voltage terms,

$$\frac{1}{2}(V_{ohm} + 2V_{stoc} + 2V_{ohm,sh})J_0 = 2eh_l n_0 u_B \varepsilon_T(T_e), \quad (\text{B7})$$

where V_{ohm} is given by

$$V_{ohm} = K_{ohm} h_l m \nu_m (l - 2s_m) \left(\frac{\omega}{e} \right)^{3/2} \left(\frac{\varepsilon_0 s_m e T_e}{J_0} \right)^{1/2}. \quad (\text{B8})$$

J_0 is the current per unit area of electrode (since our problem is 1D) and n_0 is the peak plasma density. Then, we have $K_{ohm} = 1.55$ in a collisionless (low-pressure) sheath and $K_{ohm} = 1.14 \sqrt{s_m / \lambda_i}$ in a collisional (intermediate pressure) sheath. $\nu_m = n_g \bar{\sigma}_{el} \bar{v}_e$ is the electron-neutral elastic collision frequency, with $\bar{\sigma}_{el} = 1.0 \times 10^{-19}$ and $\bar{v}_e = (8eT_e / \pi m)^{1/2}$. Next, we have,

$$V_{stoc} = K_{stoc} (meT_e)^{1/2} \left(\frac{\omega s_m}{e} \right), \quad (\text{B9})$$

where $K_{stoc} = 0.72$ in a collisionless sheath and $K_{stoc} = 0.8$ in a collisional sheath. Next,

$$V_{ohm,sh} = K_{ohm,sh} m \nu_m s_m \left(\frac{\omega s_m}{e} \right), \quad (\text{B10})$$

where $K_{ohm,sh} = 0.33$ in a collisionless sheath and $K_{ohm,sh} = 0.155$ in a collisional sheath. $\omega \equiv \omega_{RF}$ is the driving angular frequency.

As for the right-hand side of Eq. (B7) we have,

$$\varepsilon_T(T_e) = \varepsilon_{iz} + \frac{K_{exc}}{K_{iz}} \varepsilon_{exc} + \frac{3m}{M} \frac{K_{el}}{K_{iz}} T_e + 2T_e + \Delta\phi, \quad (\text{B11})$$

where

$$K_{exc} = K_{exc0} \exp\left(-\frac{\varepsilon_{exc}}{T_e}\right), \quad (\text{B12})$$

with $K_{exc0} = 0.16 \times 10^{-18} \text{1/m}^3 \text{s}$ and $\varepsilon_{exc} = 12.38 \text{ eV}$. We also have $K_{el} = \nu_m / n_g$, and finally, $\Delta\phi = 3/8 V_0$ (although $\Delta\phi = 1/2 V_0$ is a viable choice).

Writing the power balance as $n_0 = g(T_e, s_m)$ gives,

$$n_0(T_e, s_m) = \frac{(V_{ohm}(T_e, s_m) + 2V_{stoc}(T_e, s_m) + 2V_{ohm,sh}(T_e, s_m))J_0}{4eh_l u_B(T_e) \varepsilon_T(T_e)}. \quad (\text{B13})$$

3. Sheath size

The relationship between the sheath size s_m and peak plasma density n_0 depends on whether the sheath is collisionless or weakly collisional. For the collisionless case we have,

$$s_m = \frac{5}{12(eh_l n_0)^2 \varepsilon_0 T_e} \left(\frac{J_0}{\omega} \right)^3. \quad (\text{B14})$$

In the collisional case, we have,

$$s_m = 0.88 \left(\frac{\lambda_i}{\varepsilon_0 e^2 T_e \omega^3 h_l^2} \right)^{1/2} \frac{J_0^{3/2}}{n_0}. \quad (\text{B15})$$

These are re-written as

$$h(T_e, n_0, s_m) = s_m - \frac{5}{12(eh_l n_0)^2 \varepsilon_0 T_e} \left(\frac{J_0}{\omega} \right)^3, \quad (\text{B16})$$

and

$$h(T_e, n_0, s_m) = s_m - 0.88 \left(\frac{\lambda_i}{(eh_l n_0)^2 \varepsilon_0 T_e} \right)^{1/2} \left(\frac{J_0}{\omega} \right)^{3/2}. \quad (\text{B17})$$

4. Relating current to voltage amplitude

We will also need,

$$V_0 \approx \frac{s_m J_0}{K_{cap} \omega \varepsilon_0}, \quad (\text{B18})$$

where $K_{cap} = 0.613$ in a collisionless sheath and 0.751 in a collisional sheath.

This gives,

$$J_0 \approx \frac{K_{cap} \omega \varepsilon_0 V_0}{s_m}. \quad (\text{B19})$$

5. Solution algorithm

The system of Eqs. (B6), (B13), and (B16) [or Eq. (B17)] with Eq. (B19) are solved iteratively via a bisection approach. This is chosen due to the presence of small gradients in many of the functions. In the outermost loop, we iterate over Eq. (B16) or Eq. (B17) for the sheath size. At each iteration, this requires an internal iteration over the particle balance equation (B6) to compute the electron temperature. The power balance equation (B13) is used directly to predict n_0 .

To decide whether to use the collisional (intermediate pressure) or collisionless (low pressure) model, we compute the discharge properties with both models and choose that which best matches the true value from PIC simulations. This approach naturally provides more conservative estimates for the results in this paper.

APPENDIX C: SELECT CONVERGENCE PROFILES FOR CCP SIMULATIONS AT VARIOUS FREQUENCIES AND PRESSURES

Figures 10 and 11 plot the evolution of convergence [Eq. (5)] against number of simulation time steps and against number of RF

cycles, respectively, for PIC simulations of CCPs at various driving frequencies and pressures. The chosen cases represent a variety of convergence behaviors representative of that observed across the entire parameter range. This includes typical convergence ($F = 50.34$ MHz, $P = 25.13$ mTorr), rapid convergence ($F = 7.76$ MHz, $P = 24.94$ mTorr), slow monotonic convergence

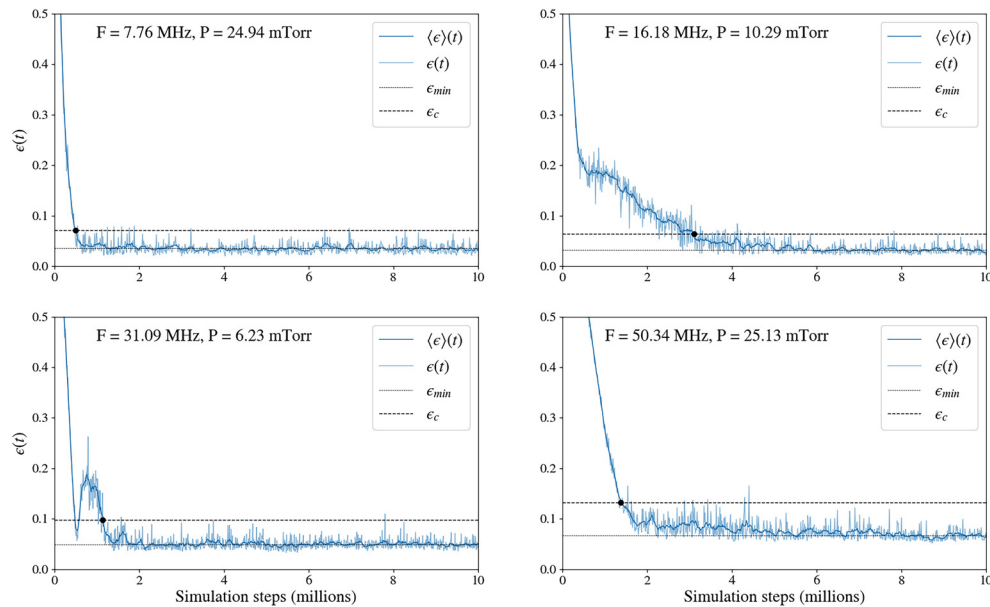


FIG. 10. Select convergence profiles vs number of simulation time steps at four different driving frequencies and pressures (values provided in figures). The black dot on each figure indicates the time at which the offline criterion measures convergence.

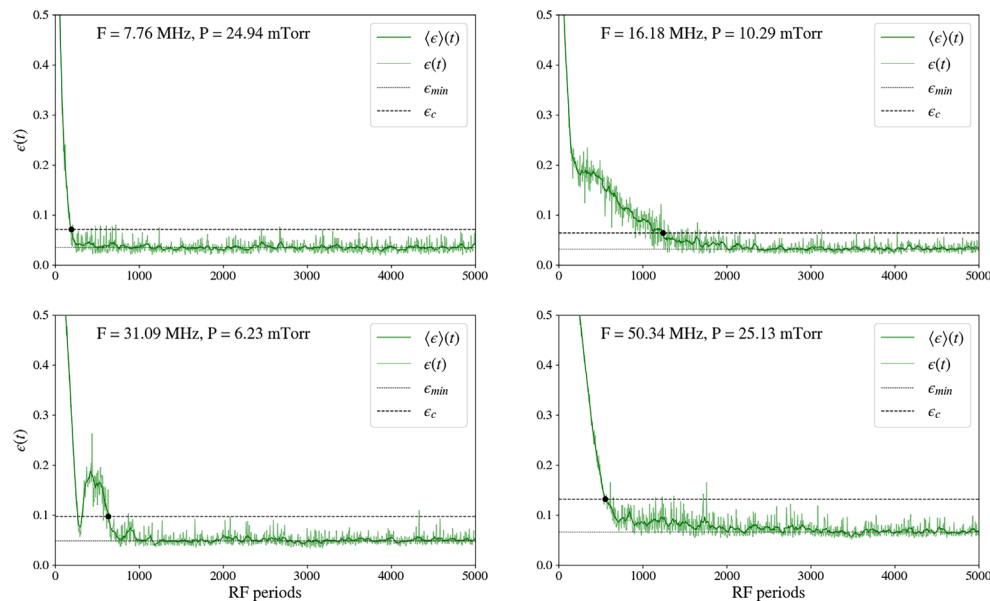


FIG. 11. Select convergence profiles vs number of RF cycles at four different driving frequencies and pressures (values provided in figures). The black dot on each figure indicates the convergence time as measured by the offline criterion.

($F = 16.18$ MHz, $P = 10.29$ mTorr), and convergence with a false minimum ($F = 31.09$ MHz, $P = 6.23$ mTorr). As shown in the figures, the procedure for measuring convergence time is robust against these different behaviors.

APPENDIX D: SELECT CCP IVDF PROFILES GENERATED FROM KINETIC PIC SIMULATIONS AND ML MODELS

Select 1D-1V IVDF profiles at four different CCP operating conditions (Fig. 12).

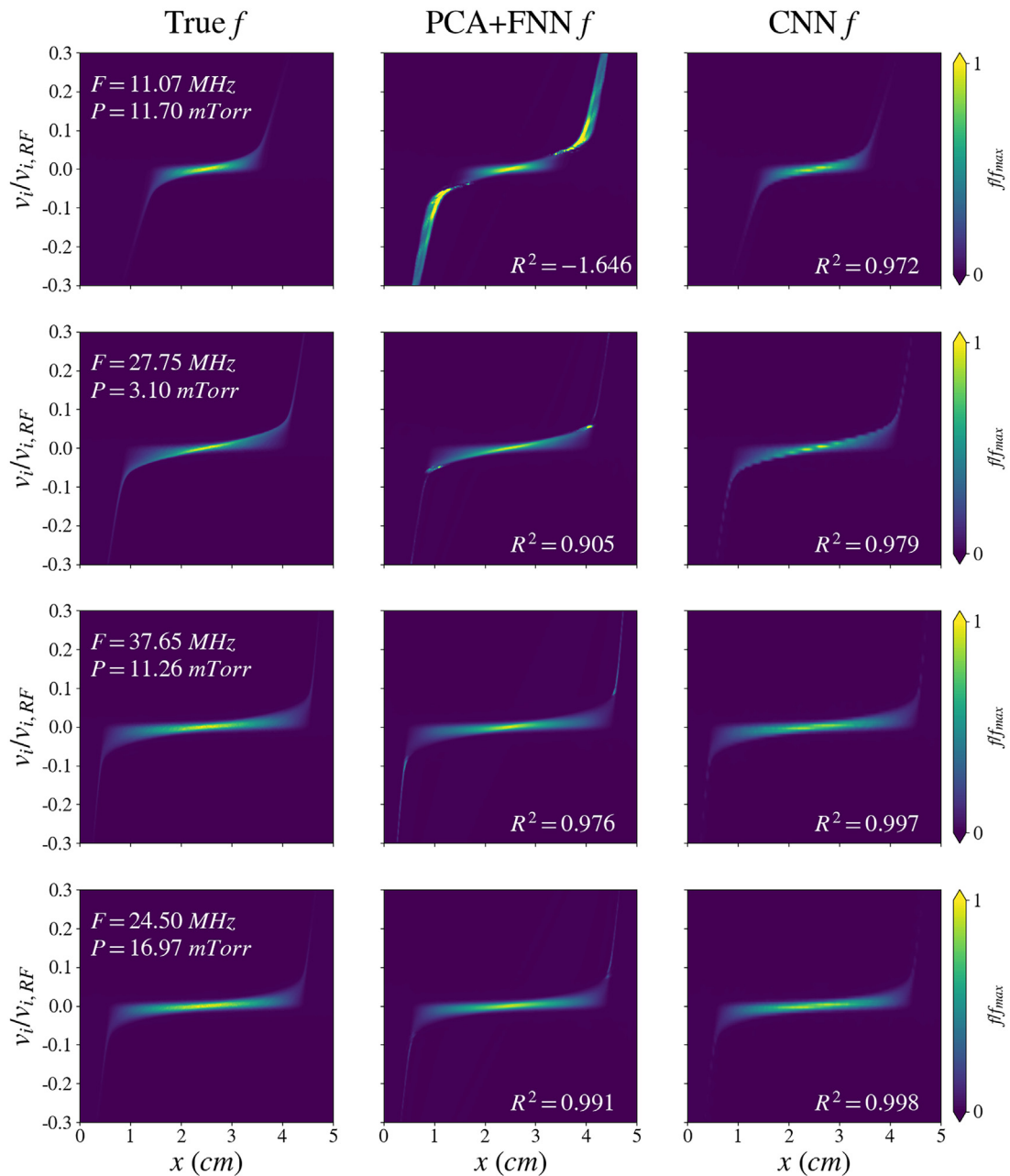


FIG. 12. Select 1D-1V IVDF profiles at four different CCP operating conditions. Each row represents a different operating condition with the gas pressure P and driving frequency F given at the top left of the figures in the left column. The left column represents the “true” IVDFs from time-averaged diagnostics of the PIC simulation after reaching quasi-steady state. The center column shows the IVDFs generated by the PCA+MLP model. The right column shows the IVDFs generated using the CNN model. R^2 values for each model are shown at the bottom right within their respective figures.

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